Paulina Czarnecki

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ACADEMIC APPOINTMENTS

Postdoctoral Researcher *November 2025 – present*
Climaviation, Sorbonne Université
- Advisors: Prof. Nicolas Bellouin, Dr. Olivier Boucher, Dr. Étienne Vignon

EDUCATION

Columbia University
Applied Mathematics – Doctor of Philosophy *August 2025*
Applied Mathematics – Masters of Science, Philosophy *February 2022, 2024*

University of Michigan, Ann Arbor *May 2020*
Honors Mathematics – Bachelor of Science
Computer Science – Minor

RESEARCH EXPERIENCE

Doctoral Research *May 2021 – August 2025*
Columbia University Applied Physics and Applied Mathematics
Lamont-Doherty Earth Observatory Ocean and Climate Physics
- Advisors: Prof. Robert Pincus and Prof. Lorenzo Polvani
- Thesis: *Computational and Theoretical Approaches to Spectral Challenges in Atmospheric Radiation*

Visiting Researcher *July – August 2023; July 2026*
Max Planck Institute for Meteorology / University of Hamburg, Hamburg, Germany
- Host: Prof. Stefan Buehler, Prof. Bjorn Stevens

Undergraduate Research *February 2017 – May 2020*
University of Michigan Biophysics
- Advisor: Prof. Michał Żochowski
- Thesis: *Effects of Cholinergic Modulation on Dynamics in Scale Free Networks*

ADDITIONAL TRAINING

WCRP km-Scale Hackathon *May 2025*
NOAA Geophysical Fluid Dynamics Laboratory / Princeton University
Project: *High Cloud Top Temperatures in km-Scale Models*

Dynamics and Data in the COVID-19 Pandemic Workshop *June 2020 – July 2020*
American Institute of Mathematics / California Institute of Technology
Project: *The Relationship between Air Quality and COVID-19*

PRESENTATIONS AND PUBLICATIONS

In Preparation

[8] **Czarnecki P.**, Bellouin, N., Boucher, O., Vignon, E. (In prep). *The Sensitivity of Embedded Contrail Radiative Forcing to the Radiative Transfer Parameterization.*

[7] **Czarnecki P.**, Brath M. (Submitted). *Data-Driven Quadrature for Longwave and Shortwave Absorption by Major Greenhouse Gases.*

Publications

[6] **Czarnecki P.**, Pincus R. (2026). *How Clear-Sky Spectral Overlap Shapes Radiation in Cloudy Atmospheres.* Journal of Climate.

[5] **Czarnecki P.**, Polvani L., Pincus R. (2025). *An Analytical Theory for Instantaneous Radiative Forcing Across Opacity Regimes.* Journal of Climate.

[4] Buehler S. A., Larsson R., Lemke O., Pfreundshuh S., Brath M., Adams I., Fox S., Roemer F., **Czarnecki P.**, Eriksson P. (2025). *The Atmospheric Radiative Transfer Simulator ARTS, Version 2.6 – Deep Python Integration.* Journal of Quantitative Spectroscopy and Radiative Transfer.

[3] **Czarnecki P.**, Pincus R., Polvani L. (2023). *Sparse, Empirically Optimized Quadrature for Broadband Radiative Calculations.* Journal of Advances in Modeling Earth Systems.

[2] Albrecht L.*, **Czarnecki P.***, Sakelaris B.* (2021). *Investigating the Relationship Between Air Quality and COVID-19 Transmission.* Journal of Data Science.

[1] **Czarnecki P.***, Lin J.*, Aton S., Zochowski M. (2021). *Dynamical mechanisms underlying scale-free network reorganization in low acetylcholine states corresponding to slow wave sleep.* Frontiers in Network Physiology.

* indicates co-first authors.

Invited Presentations

[4] **Czarnecki P.**, Brath, M., Pincus R., Polvani L. *Sparse, Empirically Optimized Quadrature for Broadband Spectral Integration.* SIAM CSE, Fort Worth, Texas. March 2025.

[3] **Czarnecki P.** *Data Driven Quadrature as a Fast, Flexible Gas Optics Scheme.* NASA GISS Seminar, New York, New York. February 2025.

[2] **Czarnecki P.**, Pincus R., Polvani L. *A Simple Analytical Model for Radiative Forcing by Optically-Thin Gases.* Equilibrium Climate Sensitivity (ECS) Symposium, Online. July 2024.

[1] **Czarnecki P.**, Pincus R., Polvani L. *Sparse, Empirically Optimized Quadrature for Radiative Fluxes and Heating Rates.* Joint Seminar, Max Planck Institute for Meteorology, Hamburg, Germany. July 2023.

Selected Presentations

[6] **Czarnecki P.**, Bellouin, N., Boucher, O., Vignon, E. *Sensitivity of Contrail Radiative Forcing to Radiative Transfer Parameterization.* EGU, Vienna, Austria. May 2026.

[5] Pincus R., **Czarnecki P.**, *How Clear-Sky Spectral Overlap Shapes Radiation in Cloudy Atmospheres.* EGU, Vienna, Austria. May 2026.

[4] **Czarnecki P.**, Pincus R., Polvani L. *A Simple Analytical Model for Instantaneous Radiative Forcing by Optically-Thin Gases.* AGU, Washington, D.C., US. December 2024.

[3] **Czarnecki P.**, Pincus R., Polvani L. *Sparse, Empirically Optimized Quadrature for Radiative Fluxes and Heating Rates.* CERES Team Meeting, NASA GISS, NYC. October 2023.

[2] **Czarnecki P.**, Pincus R., Polvani L. *Alternatives to Correlated-K Distributions for Radiative*

Transfer Calculations. International Radiation Symposium, Thessaloniki, Greece. July 2022.

[1] Albrecht L., **Czarnecki P.**, Sakelaris B. (speaker). *Investigating the Relationship Between Air Quality and COVID-19 Transmission*. Data Science Conference on COVID-19. August 2020.

Posters

[2] **Czarnecki P.**, Pincus R., Polvani L. *A Simple Model for Instantaneous Radiative Forcing by Optically-Thin Gases*. CFMIP, Boston, US. June 2024.

[1] **Czarnecki P.**, Brath M., Kluft L., Larsson R., Buehler S., Polvani L., Pincus R. *Sparse, Empirically Optimized Quadrature for Radiative Calculations in a Radiative-Convective Equilibrium Model*. American Geophysical Union, San Francisco, US. December 2023.

TEACHING

Graduate Teaching Assistant *September 2020 – September 2023*

Columbia University APAM

- Fall 2021-2023. EESCGU 4008: Intro to Atmospheric Science.
  Responsible for grading assignments and holding weekly office hours.
- Spring 2021. APMA 4300: Intro to Numerical Methods.
  Responsible for grading assignments and weekly office hours.
- Fall 2020. APMA 4200: Partial Differential Equations.
  Responsible for grading assignments and exams, and holding twice-weekly office hours.

Grader *January 2019 – May 2020*

University of Michigan Mathematics

- Fall 2019 – Winter 2020. MATH 451: Advanced Calculus.
- Winter 2019. MATH 433: Differential Geometry.

PROFESSIONAL SERVICE

Equilibrium Climate Sensitivity (ECS) Symposium *January 2026 – present*

Organizer of a monthly international virtual symposium averaging 100 attendees.

Peer Review

Journal of Climate; Journal of Quantitative Spectroscopy and Radiative Transfer; JGR: Atmospheres.

SELECTED OUTREACH

The Weather and Climate Livestream *2025 – present*

Co-organizer and speaker at a continuous, multi-day online event aimed at bringing climate and weather science to the American public and urging the restoration of federal funding; the 2025 event was covered by the New York Times and generated over 30,000 calls to Congress.

WISC Girls' Science Day Volunteer *November 2021, April 2022, 2025*

Helped with Columbia University's event, running hands-on science experiments for middle school girls.

Letters to a Pre-Scientist *September 2023 – May 2024*

Exchanged letters with a middle school student over the course of a year with the goal of making science accessible and exciting.

LDEO Open House Volunteer *October 2023*

Led hands-on experiments at Columbia University's open house.

Women in STEM at Columbia (WISC) Holistic Mentor *September 2020 – 2023*
Met with undergraduate students over the course of a year, especially discussing their paths in math or science.

Science Fridays Guest Speaker *January 2022*
Spoke to NYC Public School 205 (middle school) about my graduate research in an accessible way.

Math Circle Volunteer *September 2018 – May 2020*
Ran a weekly activity group for middle- and high-school students to interact with high level mathematics.

HONORS AND AWARDS

2024	AGU Outstanding Student Presentation Award (OSPA) in Atmospheric Science.
2023	AIM Travel Award (support for SIAM Data Science Meeting 2023).
2020	University of Michigan Honors Critical Difference Grant (support for Joint Mathematics Meetings 2020). NYU RTG funding (support for Joint Mathematics Meetings 2020).
2016	University of Michigan Regents Merit Scholarship (top graduates of Michigan high schools).